Assessing Flood Extent and Post-Disaster Socio-Economic Recovery in Feni, Bangladesh Using SAR and Night-Time Light Data

Abstract

Bangladesh's deltaic landscape is highly susceptible to severe riverine and flash flooding, necessitating rapid mapping and recovery assessment techniques. This study investigates a recent major flood event in the Feni district using cloud-based remote sensing. Sentinel-1 Synthetic Aperture Radar (SAR) imagery was utilized to detect the precise inundation extent, employing a localized backscatter threshold and topographic masking to isolate floodwaters. The study revealed that 206.63 km^2 of area was inundated by flood water during the 2024 flood event in Feni. To assess post-flood recovery, the study analyzed the relationship of Night-Time Light (NTL) radiance changes (using VIIRS data) with population density and relative wealth index across the affected areas. A linear regression analysis showed that population density negatively correlates with economic recovery (β=-0.017), while relative wealth index (rwi) positively correlates with economic recovery (β=0.35). These findings demonstrate the efficacy of combining radar and socio-economic datasets to monitor disaster resilience in vulnerable regions.

1. Introduction

The geographically low-lying nature of Bangladesh, coupled with its complex network of transboundary rivers, makes it one of the most flood-prone nations globally. The Feni district, intersected by the Muhuri river basin, frequently experiences devastating flash floods that cause extensive damage to agriculture, infrastructure, and human settlements. Traditional methods of flood mapping and damage assessment rely heavily on field surveys, which are often slow and restricted by inaccessible terrain.

Recent advancements in Earth observation offer a rapid, large-scale alternative. While optical imagery is often hindered by the heavy cloud cover present during monsoon flood events, Synthetic Aperture Radar (SAR) can penetrate clouds to provide clear views of surface inundation. Furthermore, understanding the physical extent of a flood is only the first step; assessing how communities recover is vital for resource allocation. This study aims to bridge physical damage mapping with socio-economic recovery by answering two primary objectives:

1. Delineating the precise flood extent in Feni using Sentinel-1 SAR data.
2. Evaluating post-flood recovery by performing a linear regression analysis between changes in Night-Time Light (NTL) intensity with population density and relative wealth index.

2. Methodology

2.1 Study Area

The study focuses on the Feni district (22°44'–23°17' N, 91°15'–91°35' E), a low-lying deltaic region in southeastern Bangladesh. The area is characterized by the transboundary Muhuri River basin, which channels flash floods from the upstream Indian hills of Tripura. Elevations are generally <50m, making the district a natural hydrological sink with a sharp socio-economic contrast between dense urban hubs and rural agricultural zones.

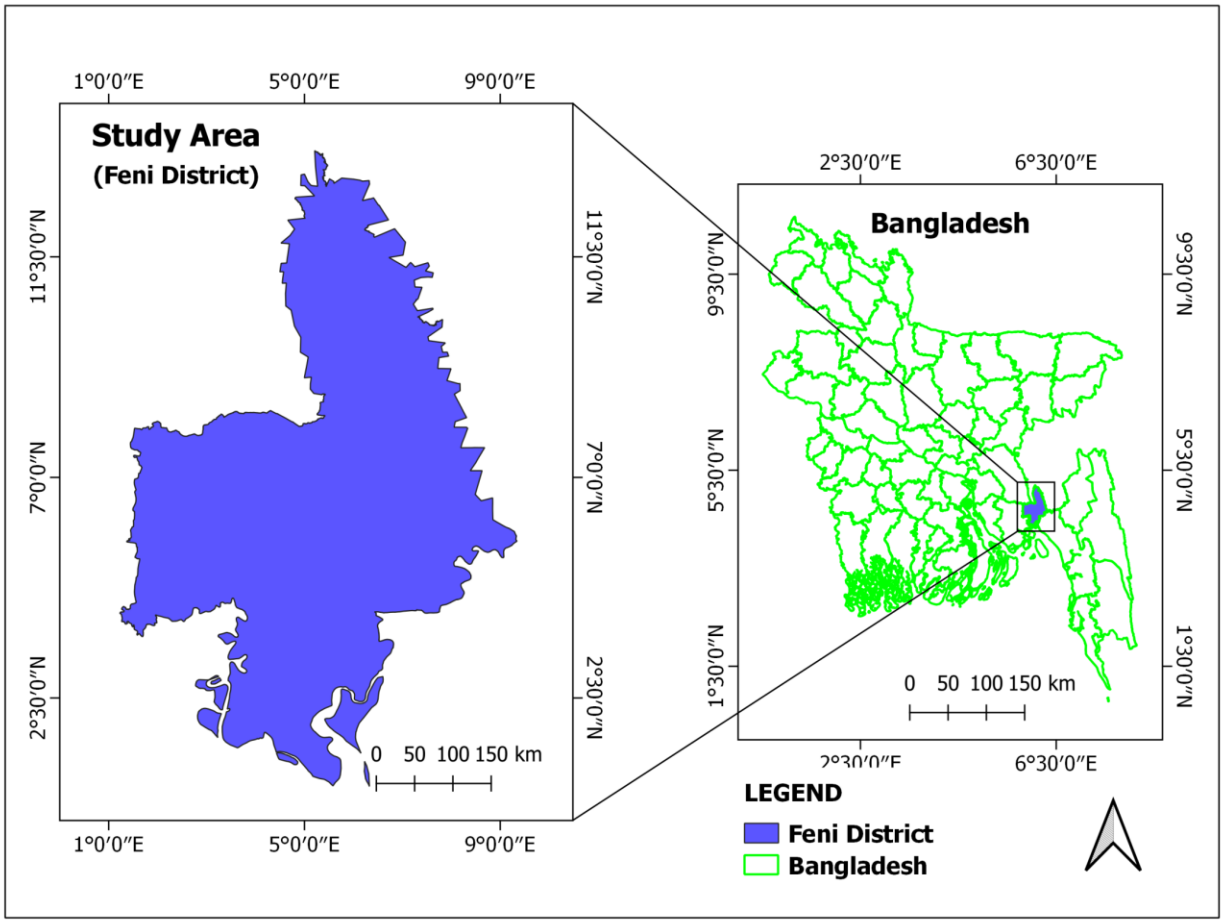

Fig-2.1: Location Map of the Study Area.

2.2 Datasets

Table 2.1: Table of datasets

Dataset	Source / Sensor	Resolution	Purpose
SAR Imagery	Sentinel-1 (C-band)	10m	Flood inundation mapping (VH polarization)
Night-time Lights	VIIRS (VNP46A2)	500m	Proxy for infrastructural and economic recovery
Relative Wealth	Meta (Data for Good)	2.4km	Estimation of economic adaptive capacity
Population	WorldPop	100m	Quantification of population density
Digital Elevation	SRTM DEM	30m	Topographic masking and shadow removal
Optical Imagery	Sentinel-2 MSI	10m	Permanent water masking (NDWI)

2.3 Flood Extent Mapping

The detection of inundated areas was performed using a multi-temporal SAR approach, supplemented by optical data for permanent water exclusion.

2.3.1 Data Acquisition & Pre-processing

Sentinel-1 Ground Range Detected (GRD) imagery (Interferometric Wide swath, VH polarization) was acquired for two distinct periods:

- **Flood Period:** August 18 – August 31, 2024.
- **Non-Flood (Baseline) Period:** January 1 – July 31, 2024.

2.3.2 Thresholding & Classification

Since water surfaces act as specular reflectors, they yield significantly lower backscatter values compared to the surrounding terrain. A binary classification was performed using a backscatter threshold of -21 dB on the flood-period VH imagery to delineate open water.

2.3.3 Accuracy Refinement & Masking

To minimize overestimation and isolate event-specific flooding, two critical masks were applied:

- **Topographic Masking:** Utilizing the SRTM 30m Digital Elevation Model, all areas with an elevation above 30m were excluded. This prevented radar shadows in the northeastern hilly fringes from being erroneously classified as water.
- **Permanent Water Masking:** A permanent water layer was generated using Sentinel-2 optical imagery from the baseline period (Jan–Mar 2024). By calculating the Normalized Difference Water Index (NDWI), existing riverine features and permanent ponds were identified and subtracted from the total flood extent.

2.4 Post-Flood Recovery Assessment

To quantify the return of human activity and infrastructural functionality, an NTL Recovery Index ($Yrec$) was formulated using VIIRS Day/Night Band radiance data.

2.4.1 Temporal Baseline Definition

Three distinct periods were established to capture the phases of the disaster:

- **Pre-flood (Baseline):** January 1 – July 31, 2024.
- **Post-flood (Impact):** August 18 – August 31, 2024.
- **Recovery (Current):** January 1 – June 30, 2025.

2.4.2 Spatial Delineation (Flood-Mask Analysis)

A critical step in this methodology was the application of a Spatial Flood Mask. The recovery analysis was performed strictly on pixels identified as inundated in Section 2.3. By clipping the NTL data to the flood extent, the study ensures that the recovery metrics reflect the actual resilience of affected communities rather than the unaffected background radiance of the district.

2.4.3 Index Calculation

The index was calculated pixel-by-pixel to measure the relative light recovery normalized against the pre-flood baseline. The formula applied was:

$$Y_{rec} = \frac{NTL_{Recovery} - NTL_{Post}}{NTL_{Pre} + 0.1}$$

A constant of 0.1 was added to the denominator to prevent mathematical undefined errors (division by zero) in dark, unlit rural pixels.

2.5 Statistical Analysis

To understand the underlying socio-economic drivers influencing the speed and scale of recovery, a multivariate linear regression was executed directly within the Google Earth Engine (GEE) environment. The model sought to determine the influence of two independent variables on the dependent NTL Recovery Index. The model is mathematically specified as:

$$Y_{rec} = \beta_0 + \beta_1(PopDensity) + \beta_2(Wealth) + \epsilon$$

Where β_0 represents the intercept, β_{1-2} represent the coefficients for Population Density and Wealth Index respectively, and ϵ accounts for the error term.

3. Results

3.1 Flood Detection Extent

The Sentinel-1 SAR analysis successfully mapped the spatial distribution of the floodwaters across Feni. The total newly inundated area was calculated to be 206.63 km^2.

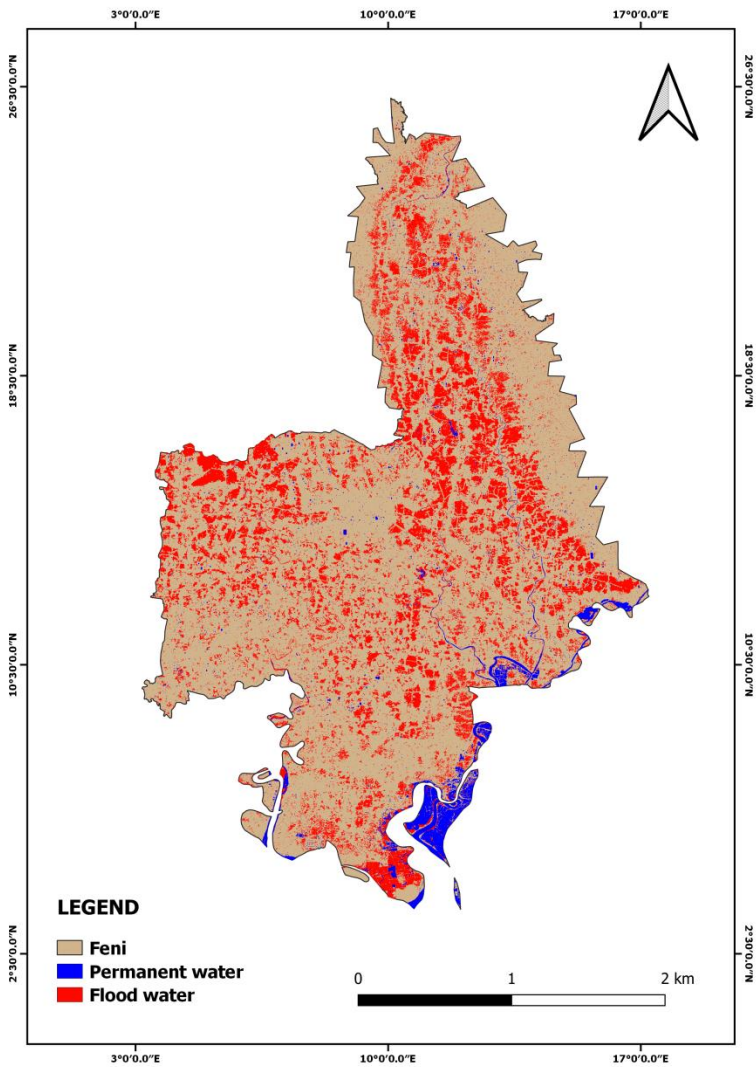

Fig-3.1 : Flood Extent Map of Feni District.

3.2 Post-Flood Recovery Regression

The multivariate regression model successfully converged, yielding empirical coefficients that describe the regional recovery dynamics across the inundated zones. The resulting equation defining the relationship is:

$$Y_{rec} = 0.64 - 0.017(PopDensity) + 0.35(Wealth)$$

The direction and magnitude of influence for each variable are detailed in Table 1 below.

Table 3.1: Regression Coefficients

Variable	Coefficient (β)	Direction of Influence
Intercept (β0)	0.64	Baseline Positive Trend
Population Density (β1)	-0.017	Weak Negative
Wealth Index (β2)	0.35	Strong Positive

The positive intercept (β=0.64) indicates that the region is experiencing a general, baseline recovery trajectory independent of the specific modeled variables.

4. Discussion

4.1 Analysis of Flood Inundation Extent

The Sentinel-1 SAR analysis successfully delineated the 2024 flash flood extent in Feni, overcoming the cloud-cover limitations of optical sensors. The -21 dB threshold effectively isolated open water via specular reflection. A critical refinement was the SRTM-based topographic mask (elevations <30m), which eliminated radar shadow noise in the northeastern hilly regions. This ensured the flood footprint was accurately confined to the low-lying Muhuri River basin, providing a precise spatial boundary for the recovery analysis.

4.2 Wealth as the Primary Driver of Recovery

The regression analysis identified the Wealth Index (β_2 = 0.35) as the strongest positive predictor of NTL recovery. This highlights a significant disparity in adaptive capacity; wealthier zones likely possess the financial liquidity to repair private electrical infrastructure or deploy decentralized power (e.g., generators/solar) immediately. In contrast, lower-income areas face a "recovery lag," as they are more dependent on the restoration of centralized state utilities, which are often delayed in marginalized zones.

4.3 The Population Density Recovery Lag

The Population Density (β_1 = -0.017) coefficient revealed a weak negative influence, suggesting that high-density areas did not recover faster. This is likely due to the infrastructural complexity of congested urban grids. The high volume of individual connections and the severity of waterlogging in built-up, impervious environments (which prevents natural drainage) likely extended the time required for electrical components to be safely reactivated compared to more permeable rural landscapes.

4.4 Synthesis of Results

Overall, the model (β_0 = 0.64) confirms a baseline positive recovery trend across Feni. However, the data proves that economic resources (β_2) are far more influential than population volume (β_1) in determining how quickly a community "lights up" after a disaster. These findings suggest that disaster management policies should prioritize infrastructural aid for high-density, low-wealth regions to mitigate prolonged post-flood outages.

5. Conclusion

This study successfully integrated remote sensing with socio-economic datasets to evaluate both the physical and economic impacts of flash flooding in Feni, Bangladesh. The customized SAR thresholding provided an accurate delineation of the disaster footprint. Furthermore, the regression analysis highlights that population density is a less significant predictor of post-disaster infrastructural recovery, as proxied by night-time light emissions. These methodologies offer scalable tools for policymakers to monitor disaster resilience and target relief efforts more effectively in the future.

6. Limitations

Several technical limitations should be acknowledged:

1. **SAR Constraints in Vegetation:** Sentinel-1 C-band radar can struggle to penetrate dense vegetation canopies (such as dense agroforestry or forested patches), potentially leading to an underestimation of floodwater hidden beneath trees.
2. **NTL Blooming and Saturation:** VIIRS data is subject to "blooming," where light from urban centers spills over into adjacent dark pixels, potentially skewing the recovery radiance values in peripheral peri-urban zones.
3. **Data Resolution:** The spatial resolution of the NTL data is coarser than the SAR imagery; meaning localized, household-level recovery dynamics cannot be captured perfectly.